SANSKRUTI NAIK

AI / ML Engineer — Generative AI · RAG · Computer Vision

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Portfolio · GitHub · LinkedIn

SUMMARY

AI/ML engineer (B.Tech AI & ML, graduated October 2026) who takes AI from notebook to production: private, citation-grounded RAG systems, explainable computer vision and secure real-time applications. Owns the full path from data and model to API, mobile client and stakeholder walkthrough. Seeking an AI/ML or GenAI engineering role on products people rely on.

EXPERIENCE

AI / ML Intern — Endureva Innovations LLP — Jan 2026 – Jun 2026

Tender Intelligence — offline RAG platform · Live app · Python · LangChain · FastAPI · Ollama · ChromaDB · SQLite FTS5 · MCP

- Designed the retrieval pipeline and LLM prompting that answer eligibility, EMD, scope, risk and penalty questions with **clause-level citations (source sentence + page)**, cutting tender analysis **from hours to minutes**.
- Combined vector and keyword search with rank fusion and **cross-encoder reranking**; measured recall, rank and latency against a golden question set.
- Built Go/No-Go, risk-finder and contradiction-finder modules, plus a Redmine integration via **MCP**; runs fully offline for confidential documents.

Crop-disease diagnosis · CNN · FastAPI · Flutter · Grad-CAM

- Shipped the feature end to end: data pipeline, CNN training, a **FastAPI** service and a **Flutter** app built for field use (**38 diseases, 92% accuracy**).
- **Under 200 ms** per prediction and **30% faster** inference after quantization and batching; **Grad-CAM** heatmaps show users which part of the leaf drove each diagnosis.

SecureChat — end-to-end encrypted messaging · React 18 · XMPP/ejabberd · OMEMO2 · WebRTC · PostgreSQL · Docker · Capacitor

- Architected **OMEMO2 (Signal Protocol)** encryption across messages, media and calls, with a local-first IndexedDB store for offline history.
- Implemented WebRTC voice/video via TURN/STUN and encrypted file sharing **up to 500 MB** using chunked AES-GCM streaming; shipped Android and iOS builds from one codebase.

Satellite construction detection · YOLOv8 · Sentinel-2 · SAR · Cartosat-3 · PostGIS · client work, confidential

- Worked on an ML system that detects unauthorised construction along a pipeline corridor (**YOLOv8** model ensemble), building its **PostGIS** spatial database, Web GIS dashboard and automated five-day alert cycle.

Offline-first apps · Flutter · FastAPI · SQLite

- **Water Survey:** a field data-collection app with geo-tagging that works without signal and auto-syncs when the network returns.
- **Restaurant POS:** role-based access, inventory alerts and live sales analytics (code); presented results and trade-offs to non-technical stakeholders.

PROJECTS

XAI-Care — Explainable Multimodal Fall Detection — 2025 – 2026
TensorFlow · FastAPI · Grad-CAM · SHAP · Flutter

- Built the multimodal fusion logic (wearable sensors + camera) and a caregiver dashboard where Grad-CAM and SHAP explain every alert.

Also built: 3D portfolio with a motion-captured avatar, Three.js (live, code) · Sign-language learning app (Smart India Hackathon)

TECHNICAL SKILLS

Generative AI: LLMs, RAG, LangChain, OpenAI API, Hugging Face, Ollama, prompt engineering, vector embeddings, ChromaDB, hybrid search, reranking, MCP

Machine Learning & Vision: TensorFlow, PyTorch, scikit-learn, CNNs, YOLOv8, OpenCV, model quantization, Grad-CAM, SHAP, NLP

Languages: Python, JavaScript, SQL, Dart, R, HTML/CSS · **Backend & Data:** FastAPI, REST APIs, PostgreSQL, PostGIS, SQLite, JWT

Data & BI: Power BI, data visualisation · **Apps & Real-time:** React, Flutter, Android, Capacitor, WebRTC, XMPP, Signal Protocol

DevOps & Tools: Docker, Docker Compose, Nginx, Git, Vercel

EDUCATION

B.Tech, Artificial Intelligence & Machine Learning — Sanjay Ghodawat University · CGPA 8.36 / 10 — Graduated Oct 2026

12th Standard, Science — Vivekanand College, Kolhapur · **Languages:** English, Hindi, Marathi (fluent), German (basic) — 2022

CERTIFICATIONS, LEADERSHIP & AWARDS

- **Certifications:** NPTEL AI for Management (12 weeks, 2026, 82/100) · Cloud & Edge Computing workshop, SGU with Prof. Rajkumar Buyya (Univ. of Melbourne), 2022 · ISRO Remote Sensing & Digital Image Analysis · Accenture Software Engineering Simulation · C++ OOP
- **3rd Prize, Smart India Hackathon** (sign-language learning app) · **Management Team Lead, Prabal Hackathon** (40 hours, 600+ participants) · **Core Member, GDG SGU**